



QSPR Analysis of Some Novel Drugs Used in Blood Cancer Treatment Via Degree Based Topological Indices and Regression Models

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ABSTRACT

Topological indices (TIs) have several biological applications in the treatment of blood cancer. TIs can be used to predict the efficacy of drugs in cancer treatment by providing information about the molecular structures of the drugs and their related properties. By analyzing the degree-based TIs of different drugs, researchers can identify the most effective drugs for treating specific types of blood cancer. Additionally, TIs can also be used to predict the toxicity of drugs, which is essential for developing safe and effective treatments. Furthermore, TIs can aid in the discovery and design of new drugs by providing insights into the relationship between molecular structure and biological activity. This article focuses on the applications of topological indices (TIs) in predicting the physical and biological properties of novel drugs used in the treatment of blood cancer. The drugs glasdegib, palbociclib, and daunorubicin are analyzed using degree-based topological indices computed through edge partitioning. A quantitative structure-property relationship (QSPR) model is then developed using linear regression to predict the properties such as boiling point (BP), flash point (FP), molar volume (MV), molecular weight, and complexity. The results demonstrate the potential of topological indices as a tool for drug discovery and design in the field of cancer treatment.

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1. Introduction

Cancer is a disease characterized by the uncontrolled growth of cells, which can originate from any part of the body, including the blood, breast and skin. It is commonly caused by DNA damage resulting from factors such as smoking and exposure to ultraviolet rays. While it was difficult to find a cure for cancer in the past, there are now several treatments available, including drug therapy. However, drug development is a costly, challenging and time-consuming process. Many drugs have been discovered for cancer treatment, including Glasdegib, Daunorubicin, and Palbociclib,¹ which are used in chemotherapy to slow down the growth of cancer cells in the body.² One of the most fatal illnesses in both humans and other organisms is cancer. Breast cancer, which occurs in the breast, is the second most common type of cancer after skin cancer. There are varying recommendations for treating the condition, as it has worsened due to a lack of appropriate treatment. In the case of breast cancer treatment, researchers have explored several newly identified entropy indices based on the degree of end vertices of each edge. To evaluate the

effectiveness of entropy indices, a QSPR was created between entropy indices and the physical characteristics of breast cancer treatments.³

Cheminformatics is a recent field that combines mathematics, information science, and chemistry to study the relationship between QSPR and QSAR, as well as the bioactivity and characterization of chemical compounds. In order to analyze networks and comprehend their underlying topology, topological indices are essential tools. Two new degrees that have just been defined in graph theory are the Ev-degree and the Ve-degree. These new graph invariants both Ev-degree and Ve-degree are parallel to the corresponding classical degree versions⁴ and reverse degree-based topological indices are also widely utilized in antibiotics.⁵ Topological indices, which are numerical measurements, are used to define the molecular topology of a graph. Topological indices are numerical values that are used to depict the molecular topology of networks in theoretical chemistry. Numerous physical and chemical properties of chemical substances can be identified through topological indices. The molecular graph theory states that a chemical compound can be depicted as a graph (or diagram), or more commonly as a molecular graph, with atoms acting as nodes and bonds between them as edges. To calculate different topological indices for these drugs, we use their chemical formulas and draw their molecular structures. Topological indices were introduced by Harry Wiener in 1947⁶ and are helpful to characterize the constructions of molecular graphs.⁷ The other distance, resistance and eccentricity based topological indices are investigated in.^{8–24} Some of the topological indices we consider in this paper are Atom-bond connectivity (ABC) index, the Randic (RA) index, Geometric-arithmetic (GA) index, Sum connectivity (S) index, First and Second Zagreb (M_1 & M_2) indices, Harmonic (H) index, Hyper Zagreb (HM) index and Forgotten (F) index. These indices provide numerical values that are obtained for the molecular formulas of the drugs.²⁵

We will use linear regression^{26,27} to compute different properties of these drugs, such as boiling point (BP), molar volume (MV), and flash point (FP) to calculate a correlation among topological indices and physicochemical characterizations. Using the resulting correlation, we will conduct QSPR model²⁸ to predict the physicochemical properties of the drugs accurately.^{29,30}

The applications of topological indices in drug development are numerous. By analyzing the structure of molecular graphs using topological indices, researchers can identify the most effective drugs³¹ for specific types of cancer, predict drug toxicity and side effects, and aid in the discovery and design of new drugs. Overall, the use of topological indices in drug development has significant potential to improve cancer treatment and enhance our understanding of molecular structures. Some degree based topological indices of two novel metal–organic models are obtained in³² and other topological indices for some chemical graphs are investigated in.^{33–41}

The molecular structure of drugs is represented as a graph where the atoms are the vertices and the bonds connecting them are the edges. The graph $G(V, E)$ is a simple, finite, and connected representation of the drug's structure, where V and E are the sets of vertices and edges, respectively. In graph theory, the degree of any node in a given graph, denoted as d_u , refers to the amount of nodes that are adjacent to it. The valence of a compound in chemistry is equivalent to the degree of its corresponding node in the graph. The table below presents a list of topological indices, along with their corresponding notations, formulas, and year of invention (Table 1).

2. Results and discussions

Molecular descriptors have a huge application in the field of medicine, particularly in drug design and discovery. In the case of blood cancer treatment, topological indices can be used to identify potential drug candidates with the desired physicochemical properties. By imposing degree-based TIs on drugs used in the treatment of blood cancer, we can gain insights into their structural characteristics and relate them to their biological activity.

Table 1. The notations, formulas and year of invention of topological indices.

Topological index	Notation	Formula	Reference
ABC Index	ABC(G)	$\sum_{uv \in E(G)} \sqrt{\frac{du+dv-2}{du \times dv}}$	Ernesto Estrade, 1998 ⁴²
Randic Index	RA(G)	$\sum_{uv \in E(G)} \sqrt{\frac{1}{du \times dv}}$	In 1975, Million Randic ⁴³
Sum Connectivity	S(G)	$\sum_{uv \in E(G)} \sqrt{\frac{1}{du+dv}}$	Zhou and Trinajstić ⁴⁴
Geometric-arithmetic index	GA(G)	$\sum_{uv \in E(G)} 2 \sqrt{\frac{du \times dv}{du+dv}}$	Shegehall and Kanabur ⁴⁵
First Zagreb Index	M ₁ (G)	$\sum_{uv \in E(G)} (du + dv)$	Gutman and Trinajstić ⁴⁶
Second Zagreb Index	M ₂ (G)	$\sum_{uv \in E(G)} (du \times dv)$	
Harmonic index	H(G)	$\sum_{uv \in E(G)} \frac{2}{du+dv}$	Graffiti ⁴⁷
Hyper Zagreb index	HM(G)	$\sum_{uv \in E(G)} (du + dv)^2$	G. Shirdel, H. Rezapour and A. Sayadi ⁴⁸
Forgotten index	F(G)	$\sum_{uv \in E(G)} [(du)^2 + (dv)^2]$	Furtula and Gutman 2015 ⁴⁹

The QSPR modeling technique is useful for examining the correlation between molecular descriptors and the physicochemical properties of drugs used in the treatment of blood cancer. By leveraging this analysis, it becomes possible to estimate the physicochemical properties of new drug candidates based on their structural features, thereby facilitating more efficient drug discovery efforts.

In this study, several drugs used in the treatment of blood cancer were analyzed using topological indices and QSPR modeling.^{50,51} The drugs included Glasdegib, Palbociclib, Daunorubicin, Gilteritinib, and Ivosieninb, and their molecular structure are shown in Figure 1. By using the degree-based TIs on these drugs, we are able to calculate their numerical values and relate them to their physicochemical properties.

The results of current analysis can have important implications in the development of new drugs for blood cancer treatment. By identifying the structural features and physicochemical properties of successful drugs, researchers can use this knowledge to design new drugs with similar properties and potentially improved efficacy. Additionally, this approach can help in the optimization of existing drugs by modifying their structural features to improve their physicochemical properties and increase their effectiveness in treating blood cancer. Regression analysis is thus applied in the study's calculations.

2.1. Regression model

Regression analysis is a powerful statistical technique that allows us to explore the relationships between two or more relevant variables. In the context of blood cancer treatment, the regression model is employed to identify correlations among the molecular descriptors and physicochemical properties of drugs. The findings of the analysis reveal a strong association between the topological indices and the physicochemical properties of the drugs.

The applications of topological indices in blood cancer are numerous. Topological descriptors are useful to analyze the structure of various drugs used in the treatment of blood cancer, including leukemia and lymphoma. The analysis of topological indices in drug design can help identify new drugs or optimize existing ones. For example, molecular descriptors gives to predict the activity of new drugs in the treatment of leukemia by analyzing the structural properties of the drugs. Additionally, topological indices can be used to determine the mode of action of drugs in the treatment of blood cancer, which can provide valuable insights into the biological mechanisms of these drugs. Overall, the use of topological indices in the study of blood cancer can lead to the discovery of new drugs and improve the effectiveness of existing ones. Our obtained results

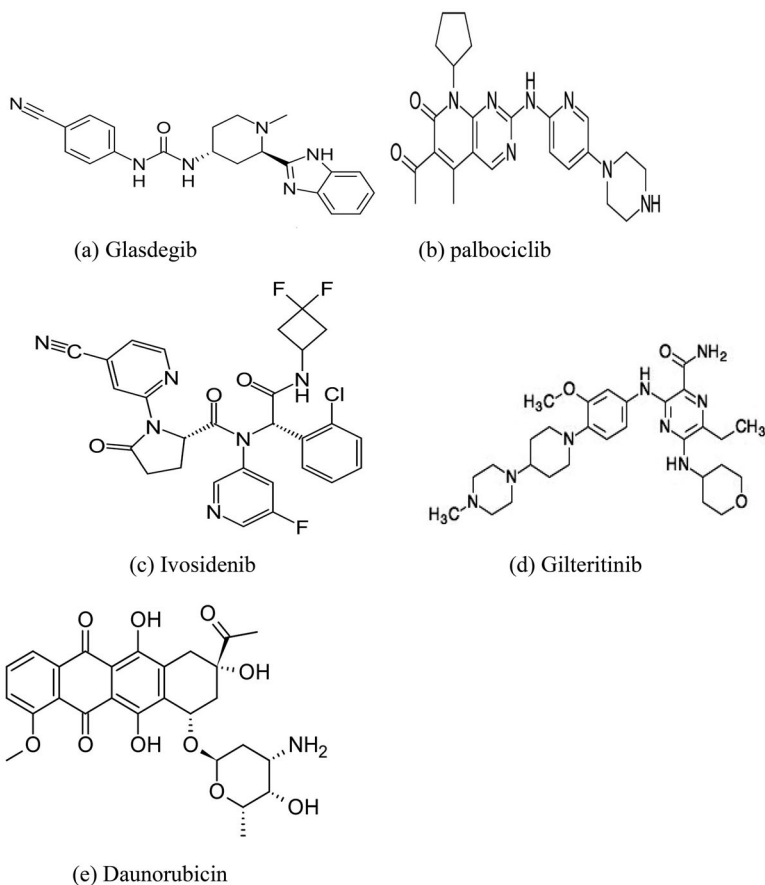


Figure 1. Molecular structure of drugs.

are tested with [Equation \(1\)](#).

$$P = A + b(TI) \quad (1)$$

In this context, the symbol P denotes a parameter related to the physicochemical properties of a potential drug. Additionally, TI represents a topological index, while A and b correspond to the regression coefficients used in the analysis.

Theorem 1. Let G_1 denotes the Glasdegib graph, then the following results holds

- i. $ABC(G_1) = 39.16$
- ii. $RA(G_1) = 22.47$
- iii. $S(G_1) = 23.08$
- iv. $GA(G_1) = 49.02$
- v. $M1(G_1) = 290$
- vi. $M2(G_1) = 377$
- vii. $H(G_1) = 20.36$
- viii. $HM(G_1) = 1664$
- ix. $F(G_1) = 910$

Proof. Let G_1 belongs to Glasdegib, having edge set E , and E_m , denotes the set of edges in G_1 adding vertices of degrees m and n . The frequencies are given as $|E_{1,2}| = 1$, $|E_{1,3}| = 12$, $|E_{1,4}| = 11$, $|E_{2,3}| = 3$, $|E_{3,3}| = 17$, $|E_{3,4}| = 5$, $|E_{4,4}| = 4$. Then,

- i. By using ABC index and above given edge partitions $E_{m,n}$ we get

$$ABC(G) = \sum_{uv \in E(G)} \sqrt{\frac{du+dv-2}{dudv}}$$

$$ABC(G_1) = 1\sqrt{\frac{1+2-2}{1 \times 2}} + 12\sqrt{\frac{1+3-2}{1 \times 3}} + 11\sqrt{\frac{1+4-2}{1 \times 4}} + 3\sqrt{\frac{2+3-2}{2 \times 3}} + 17\sqrt{\frac{3+3-2}{3 \times 3}} + 5\sqrt{\frac{3+4-2}{3 \times 4}} + 4\sqrt{\frac{4+4-2}{4 \times 4}} = 39.16$$

- ii. By using Randic index and edge partitions $E_{m,n}$ one has,

$$RA(G) = \sum_{uv \in E(G)} \sqrt{\frac{1}{du \times dv}}$$

$$RA(G_1) = 1\sqrt{\frac{1}{1 \times 2}} + 12\sqrt{\frac{1}{1 \times 3}} + 11\sqrt{\frac{1}{1 \times 4}} + 3\sqrt{\frac{1}{2 \times 3}} + 17\sqrt{\frac{1}{3 \times 3}} + 5\sqrt{\frac{1}{3 \times 4}} + 4\sqrt{\frac{1}{4 \times 4}} = 22.47$$

By using Sum connectivity index and edge partitions $E_{m,n}$ one has,

$$S(G) = \sum_{uv \in E(G)} \sqrt{\frac{1}{du + dv}}$$

$$S(G_1) = 1\sqrt{\frac{1}{1+2}} + 12\sqrt{\frac{1}{1+3}} + 11\sqrt{\frac{1}{1+4}} + 3\sqrt{\frac{1}{2+3}} + 17\sqrt{\frac{1}{3+3}} + 5\sqrt{\frac{1}{3+4}} + 4\sqrt{\frac{1}{4+4}} = 23.08$$

By using geometric index and edge partitions $E_{m,n}$ one has,

$$GA(G) = \sum_{uv \in E(G)} \frac{2\sqrt{dudv}}{du + dv}$$

$$GA(G_1) = 2\frac{2\sqrt{1 \times 2}}{1+2} + 12\frac{2\sqrt{1 \times 3}}{1+3} + 11\frac{2\sqrt{1 \times 4}}{1+4} + 3\frac{2\sqrt{2 \times 3}}{2+3} + 17\frac{2\sqrt{3 \times 3}}{3+3} + 5\frac{2\sqrt{3 \times 4}}{3+4} + 4\frac{2\sqrt{4 \times 4}}{4+4} = 49.02$$

By using First Zagreb index and edge partitions $E_{m,n}$ one has,

$$M_1 = \sum_{uv \in E(G)} (du + dv)$$

$$M_1(G) = 1(1 + 2) + 12(1 + 3) + 11(1 + 4) + 3(2 + 3) + 17(3 + 3) + 5(3 + 4) + 4(4 + 4) = 290$$

By using Second Zagreb index and an edge partitions $E_{m,n}$ one has,

$$M_2 = \sum_{uv \in E(G)} (du \times dv)$$

$$M_2(G_1) = 1(1 \times 2) + 12(1 \times 3) + 11(1 \times 4) + 3(2 \times 3) + 17(3 \times 3) + 5(3 \times 4) + 4(4 \times 4) = 377$$

By using Harmonic index and an edge partitions $E_{m,n}$ one has,

$$H(G) = \sum_{uv \in E(G)} \frac{2}{du + dv}$$

$$\begin{aligned} H(G_1) &= 1\left(\frac{2}{1+2}\right) + 12\left(\frac{2}{1+3}\right) + 11\left(\frac{2}{1+4}\right) + 3\left(\frac{2}{2+3}\right) + 17\left(\frac{2}{3+3}\right) + 5\left(\frac{2}{3+4}\right) \\ &\quad + 4\left(\frac{2}{4+4}\right) = 20.36 \end{aligned}$$

By using Hyper Zagreb index and above given edge partitions $E_{m,n}$ we get,

$$HM(G) = \sum_{uv \in E(G)} (du + dv)^2$$

$$\begin{aligned} HM(G_1) &= 1(1 + 2)^2 + 12(1 + 3)^2 + 11(1 + 4)^2 + 3(2 + 3)^2 + 17(3 + 3)^2 + 5(3 + 4)^2 \\ &\quad + 4(4 + 4)^2 = 1664 \end{aligned}$$

By using forgotten index and an edge partitions $E_{m,n}$ one has,

$$F(G) = \sum_{uv \in E(G)} [(du)^2 + (dv)^2]$$

$$\begin{aligned} F(G_1) &= 1(1^2 + 2^2) + 12(1^2 + 3^2) + 11(1^2 + 4^2) + 3(2^2 + 3^2) + 17(3^2 + 3^2) + 5(3^2 + 4^2) \\ &\quad + 4(4^2 + 4^2) = 910 \end{aligned}$$

Theorem 2. Let G_2 denotes the palbociclib graph, then the following results holds.

- i. $ABC(G_2) = 49.49$
- ii. $RA(G_2) = 27.33$
- iii. $S(G_2) = 28.21$
- iv. $GA(G_2) = 60.13$
- v. $M_1(G_2) = 372$
- vi. $M_2(G_2) = 483$
- vii. $H(G_2) = 24.35$
- viii. $HM(G_2) = 2184$
- ix. $F(G_2) = 1218$

Proof. Let G_2 belongs to palbociclib having edge set E' , and $E'_{(m,n)}$ denotes the set of links in G_2 by adding vertices of degrees m and n . Let the frequencies of each edge is $|E'_{(1,3)}| = 8$, $|E'_{(1,4)}| = 23$, $|E'_{(2,3)}| = 6$, $|E'_{(3,3)}| = 15$, $|E'_{(3,4)}| = 7$, $|E'_{(4,4)}| = 7$ then

- i. By using ABC index and the frequency of $E'_{(m,n)}$ one has,

$$ABC(G) = \sum_{uv \in E(G)} \sqrt{\frac{du + dv - 2}{dudv}}$$

$$ABC(G_2) = 8\sqrt{\frac{1+3-2}{1 \times 3}} + 23\sqrt{\frac{1+4-2}{1 \times 4}} + 6\sqrt{\frac{2+3-2}{2 \times 3}} + 15\sqrt{\frac{3+3-2}{3 \times 3}} \\ + 7\sqrt{\frac{3+4-2}{3 \times 4}} + 7\sqrt{\frac{4+4-2}{4 \times 4}} = 49.49$$

- By using Randic index and the frequency of $E'_{(m,n)}$ one has,

$$RA(G) = \sum_{uv \in E(G)} \sqrt{\frac{1}{du \times dv}}$$

$$RA(G_2) = 8\sqrt{\frac{1}{1 \times 3}} + 23\sqrt{\frac{1}{1 \times 4}} + 6\sqrt{\frac{1}{2 \times 3}} + 15\sqrt{\frac{1}{3 \times 3}} + 7\sqrt{\frac{1}{3 \times 4}} + 7\sqrt{\frac{1}{4 \times 4}} = 27.33$$

- Using Sum connectivity index and the frequency of $E'_{(m,n)}$ one has

$$S(G) = \sum_{uv \in E(G)} \sqrt{\frac{1}{du + dv}}$$

$$S(G_2) = 8\sqrt{\frac{1}{1+3}} + 23\sqrt{\frac{1}{1+4}} + 6\sqrt{\frac{1}{2+3}} + 15\sqrt{\frac{1}{3+3}} + 7\sqrt{\frac{1}{3+4}} + 7\sqrt{\frac{1}{4+4}} = 28.21$$

- Using geometric index and the frequency of $E'_{(m,n)}$ one has,

Table 2. The molecular descriptors for the candidate drugs.

Name of drug	ABC	RA	S	GA	M ₁	M ₂	F	H	HM
Glasdegib	39.16	22.47	23.08	49.02	290	377	910	20.36	1664
Palbociclib	49.49	27.33	28.21	60.13	372	483	1218	24.35	2184
Daunorubicin	52.54	29.92	30.70	65.19	396	512	1288	26.96	2324
Gilteritinib	66.75	36.52	37.39	79.15	518	648	1688	32.07	2984
Ivosideni	49.42	28.33	29.15	62.02	379	483	990	25.7	2126

$$GA(G) = \sum_{uv \in E(G)} \frac{2\sqrt{dudv}}{du + dv}$$

$$GA(G_2) = 8 \frac{2\sqrt{1 \times 3}}{1+3} + 23 \frac{2\sqrt{1 \times 4}}{1+4} + 6 \frac{2\sqrt{2 \times 3}}{2+3} + 15 \frac{2\sqrt{3 \times 3}}{3+3} + 7 \frac{2\sqrt{3 \times 4}}{3+4} + 7 \frac{2\sqrt{4 \times 4}}{4+4} = 60.13$$

Using First Zagreb index and the frequency of $E'_{(m,n)}$ one has,

$$M_1 = \sum_{uv \in E(G)} (du + dv)$$

$$M_1(G_2) = 8(1+3) + 23(1+4) + 6(2+3) + 15(3+3) + 7(3+4) + 7(4+4) = 372$$

Using Second Zagreb and the frequency of $E'_{(m,n)}$ one has,

$$M_2 = \sum_{uv \in E(G)} (du \times dv)$$

$$M_2(G_2) = 8(1 \times 3) + 23(1 \times 4) + 6(2 \times 3) + 15(3 \times 3) + 7(3 \times 4) + 7(4 \times 4) = 483$$

Using an Harmonic index edge partition $E'_{(m,n)}$ we get,

$$H(G) = \sum_{uv \in E(G)} \frac{2}{du + dv}$$

$$H(G_2) = 8 \frac{2}{1+3} + 23 \frac{2}{1+4} + 6 \frac{2}{2+3} + 15 \frac{2}{3+3} + 7 \frac{2}{3+4} + 7 \frac{2}{4+4} = 24.35$$

Using Hyper Zagreb index and edge partition $E'_{(m,n)}$ we get,

$$HM(G) = \sum_{uv \in E(G)} (du + dv)^2$$

$$HM(G_2) = 8(1+3)^2 + 23(1+4)^2 + 6(2+3)^2 + 15(3+3)^2 + 7(3+4)^2 + 7(4+4)^2 = 2184$$

By using forgotten index and edge partition $E'_{(m,n)}$ we get,

$$F(G) = \sum_{uv \in E(G)} [(du)^2 + (dv)^2]$$

$$F(G_2) = 8(1^2 + 3^2) + 23(1^2 + 4^2) + 6(2^2 + 3^2) + 15(3^2 + 3^2) + 7(3^2 + 4^2) + 7(4^2 + 4^2) = 1218$$

The topological indices of other drugs can be obtained using a similar technique as that used in [Theorems 1](#) and [2](#), but with different topological indices. [Table 2](#) also includes a list of all medicines' computed values.

To ensure precise results, all data tables are generated using Microsoft Excel. A linear QSPR model is utilized to analyze the nine topological indices (TIs) and their corresponding physical characteristics in potential treatments for blood cancer. By employing [Equation \(1\)](#), a linear regression model is developed for the degree-based TIs of the potential drugs, resulting in the determination of A and b values for the following nine topological indices.

2.1.1. Regression models of ABC (G)

Boiling point = $666.9077 + 1.288318[\text{ABC (G)}]$
Molecular weight = $194.852 + 5.870532 [\text{ABC (G)}]$
Flash Point = $356.923 + 0.781725 [\text{ABC (G)}]$
Molar Volume = $65.81342 + 5.675058[\text{ABC (G)}]$
Complexity = $571.6181 + 5.078138 [\text{ABC (G)}]$

2.1.2. Regression models of RA (G)

Boiling point = $630.2426 + 3.561506[\text{RA (G)}]$
Molecular weight = $140.6155 + 191.2503[\text{RA (G)}]$
Flash Point = $334.7966 + 2.156858[\text{RA (G)}]$
Molar Volume = $35.59021 + 11.14788 [\text{RA (G)}]$
Complexity = $479.6946 + 12.21918 [\text{RA (G)}]$

2.1.3. Regression models of S (G)

Boiling point = $626.0781 + 3.606742[\text{S (G)}]$
Molecular weight = $135.2184 + 12.17941 [\text{S (G)}]$
Flash Point = $332.2831 + 2.183967 [\text{S (G)}]$
Molar Volume = $31.21306 + 10.99801[\text{S (G)}]$
Complexity = $470.1911 + 12.21332[\text{S (G)}]$

2.1.4. Regression models of GA (G)

Boiling point = $621.4113 + 1.771872[\text{GA (G)}]$
Molecular weight = $129.5672 + 5.823156 [\text{GA (G)}]$
Flash Point = $329.4663 + 1.072767 [\text{GA (G)}]$
Molar Volume = $27.2797 + 5.239775[\text{GA (G)}]$
Complexity = $459.3878 + 5.920767 [\text{GA (G)}]$

2.1.5. Regression models of M_1 (G)

Boiling point = $669.2816 + 0.165815[M_1 (G)]$
Molecular weight = $204.3029 + 0.759121[M_1 (G)]$
Flash Point = $358.3484 + 0.100653[M_1 (G)]$

Table 3. The properties of drugs related to their physical characteristics.

Name of drug	Molar volume (cm ³)	Boiling point °C	Molecular weight	Complexity	Flash point °C
Glasdegib	281	633.4	374.4	595.00	336.9
Palbociclib	340.7	711.5	447.5	775.00	384.1
Daunorubicin	339.4	770	527.5	960.00	419.5
Gilteritinib	444.9	696.9	552.7	785.00	375.3
Ivosidenib	383.6	854.3	583.0	1050.00	470

Table 4. Correlation coefficients.

Topological index	Correlation coefficient of molar volume	Correlation coefficient of flash point	Correlation coefficient of boiling point	Correlation coefficient of molar weight	Correlation coefficient of complexity
ABC	0.926871	0.154572	0.153516	0.685581	0.2846
RA	0.931662	0.218231	0.217161	0.736603	0.366424
S	0.933468	0.22419	0.223348	0.73917	0.355713
GA	0.934547	0.231645	0.23057	0.73917	0.362366
M ₁	0.919862	0.152966	0.151862	0.681373	0.285843
M ₂	0.923228	0.182184	0.181067	0.702272	0.315296
HM	0.911082	0.133305	0.132169	0.664348	0.268345
F	0.773686	0.110239	0.11165	0.447685	0.046966
H	0.933046	0.267344	0.266268	0.770904	0.399409

$$\text{Molar Volume} = 75.35483 + 0.732794 [M_1 (G)]$$

$$\text{Complexity} = 577.1165 + 0.663598[M_1 (G)]$$

2.1.6. Regression models of M₂ (G)

$$\text{Boiling point} = 655.4674 + 0.154947[M_2 (G)]$$

$$\text{Molecular weight} = 189.3174 + 0.613198 [M_2 (G)]$$

$$\text{Flash Point} = 350.0147 + 0.093952[M_2 (G)]$$

$$\text{Molar Volume} = 68.67423 + 0.576416[M_2 (G)]$$

$$\text{Complexity} = 545.1308 + 0.573673[M_2 (G)]$$

2.1.7. Regression models of HM (G)

$$\text{Boiling point} = 681.0432 + 0.02314[HM (G)]$$

$$\text{Molecular weight} = 229.4156 + 0.118598[HM (G)]$$

$$\text{Flash Point} = 365.4463 + 0.014055[HM (G)]$$

$$\text{Molar Volume} = 95.50563 + 0.116298[HM (G)]$$

$$\text{Complexity} = 607.7618 + 0.099822 [HM (G)]$$

2.1.8. Regression models of F (G)

$$\text{Boiling point} = 770.3666 + (-0.03048) [F (G)]$$

$$\text{Molecular weight} = 345.0425 + 0.124694 [F (G)]$$

$$\text{Flash Point} = 419.2627 + (-0.01813) [F (G)]$$

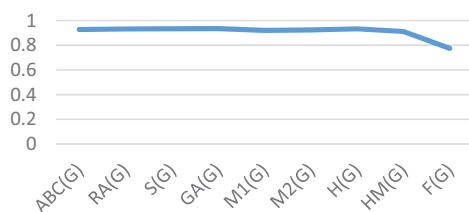
$$\text{Molar Volume} = 170.1161 + 0.154089[F (G)]$$

$$\text{Complexity} = 799.9965 + 0.027259[F (G)]$$

2.1.9. Regression models of H (G)

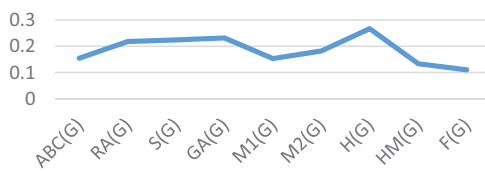
$$\text{Boiling point} = 598.1548 + 5.217292[H (G)]$$

Correlation Coefficient Of Molar Volume



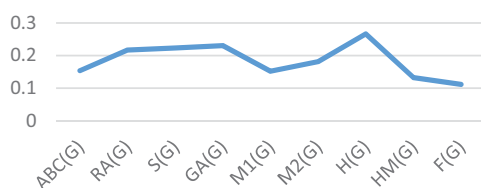
(i) Molar Volume on TI

Correlation Coefficient Of Flash Point



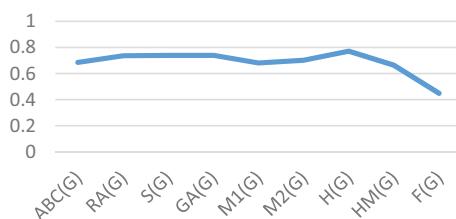
(ii) Flash Point on TI

Correlation Coefficient Of Boiling Point



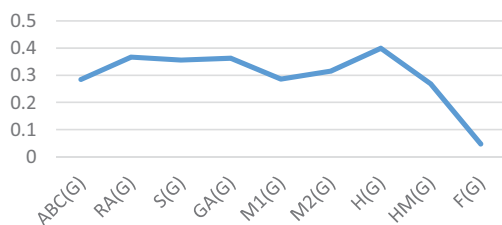
(iii) Boiling Point on TI

Correlation Coefficient Of Molar Weight



(iv) Molar Weight on TI

Correlation Coefficient Of Complexity



(v) Complexity on TI

Figure 2. Physicochemical properties and TIs.

Table 5. The statistical parameters employed in the QSPR model of ABC.

	PP							
	<i>N</i>	<i>A</i>	<i>b</i>	<i>r</i>	<i>r</i> ²	<i>F</i>	<i>p</i>	Indicator
BP	5	666.9077	1.288318	0.153516	0.023567	0.072408	0.80	Non-signi
MW	5	194.852	5.870532	0.685581	0.470022	2.660609	0.20	Non-signi
FP	5	356.923	0.781725	0.154572	0.023893	0.073432	0.80	Non-signi
MV	5	65.81342	5.675058	0.926871	0.85909	18.29014	0.02	Significant
C	5	571.6181	5.078138	0.2846	0.080997	0.264407	0.64	Non-signi

Table 6. The statistical parameters employed in the QSPR model of RA.

PP								
	<i>N</i>	<i>A</i>	<i>b</i>	<i>r</i>	<i>r</i> ²	<i>F</i>	<i>p</i>	Indicator
BP	5	630.2426	3.561506	0.21716	0.047159	0.148479	0.72	Non-signi
MW	5	140.6155	191.2503	0.736603	0.542585	30.55859	0.15	Non-signi
FP	5	334.7966	2.156858	0.218231	0.047625	0.150019	0.72	Non-signi
MV	5	35.59021	11.14788	0.931662	0.867995	190.7263	0.02	Significant
C	5	479.6946	12.21918	0.35042	0.122794	0.419951	0.56	Non-signi

Table 7. The statistical parameters employed in the QSPR model of S.

PP								
	<i>N</i>	<i>A</i>	<i>b</i>	<i>r</i>	<i>r</i> ²	<i>F</i>	<i>p</i>	Indicator
BP	5	626.0781	3.606742	0.223348	0.04988	0.15751	0.71	Non-signi
MW	5	12.17941	12.17941	0.73917	0.54637	3.613346	0.15	Non-signi
FP	5	332.2831	2.183967	0.224419	0.05036	0.159105	0.71	Non-signi
MV	5	31.21306	10.99801	0.933468	0.871363	20.32139	0.02	Significant
Complexity	5	470.1911	12.21332	0.355713	0.126532	0.434584	0.55	Non-signi

Table 8. The statistical parameters employed in the QSPR model of GA.

PP								
	<i>N</i>	<i>A</i>	<i>b</i>	<i>r</i>	<i>r</i> ²	<i>F</i>	<i>p</i>	Indicator
BP	5	621.4113	1.771872	0.23057	0.053163	93.61597	0.70	Non-Signi
MW	5	129.5672	5.823156	0.742642	0.551517	65.74071	0.15	Non-Signi
FP	5	329.4663	1.072767	0.231645	0.053659	0.170106	0.70	Non-Signi
MV	5	27.2797	5.239775	0.934547	0.873379	20.69269	0.01	Significant
C	5	459.3878	5.920767	0.362366	0.131309	0.453473	0.54	Non-Signi

Table 9. The statistical parameters employed in the QSPR model of M₁.

PP								
	<i>N</i>	<i>A</i>	<i>b</i>	<i>r</i>	<i>r</i> ²	<i>F</i>	<i>p</i>	Indicator
BP	5	669.2816	0.1651815	0.151862	0.023062	0.070819	0.80	Non-signi
MW	5	204.3029	0.759121	0.681373	0.464269	2.599825	0.20	Non-signi
FP	5	358.3484	0.100653	0.152966	0.023399	0.071878	0.80	Non-signi
MV	5	75.35483	0.732794	0.919862	0.846147	16.49911	0.02	Significant
C	5	577.1165	0.663598	0.285843	0.081706	0.266928	0.64	Non-signi

Table 10. The statistical parameters employed in the QSPR model of M₂.

PP								
	<i>N</i>	<i>A</i>	<i>b</i>	<i>r</i>	<i>r</i> ²	<i>F</i>	<i>p</i>	Indicator
BP	5	655.4674	0.154947	0.181067	0.023062	0.101689	0.77	Non-signi
MW	5	189.3174	0.613198	0.702272	0.464269	2.919338	0.18	Non-signi
FP	5	350.0147	0.093952	0.182184	0.023399	0.102991	0.76	Non-signi
MV	5	68.67423	0.576416	0.923228	0.846147	17.31832	0.02	Significant
C	5	545.1308	0.573673	0.315296	0.081706	0.102991	0.60	Non-signi

Table 11. The statistical parameters employed in the QSPR model of HM.

PP								
	<i>N</i>	<i>A</i>	<i>b</i>	<i>r</i>	<i>r</i> ²	<i>F</i>	<i>p</i>	Indicator
BP	5	681.0432	0.023124	0.132169	0.017469	0.053337	0.83	Non-signi
MW	5	229.4156	0.118598	0.664348	0.441359	2.370173	0.22	Non-signi
FP	5	365.4463	0.014055	0.133305	0.01777	0.054275	0.83	Non-signi
MV	5	95.50563	0.116298	0.911082	0.830071	14.65444	0.03	Significant
C	5	607.7618	0.099822	0.268345	0.072009	0.23279	0.66	Non-signi

Table 12. The statistical parameters employed in the QSPR model of H.

PP								
	<i>N</i>	<i>A</i>	<i>b</i>	<i>r</i>	<i>r</i> ²	<i>F</i>	<i>p</i>	Indicator
BP	5	598.1548	5.217292	0.266268	0.070899	0.228927	0.66	Non-signi
MW	5	98.01885	15.41259	0.770904	0.0594293	4.394502	0.12	Non-signi
FP	5	315.4363	3.156817	0.267344	0.071473	0.230923	0.66	Non-signi
MV	5	12.60989	13.33862	0.933046	0.870575	20.17937	0.02	Significant
C	5	402.2318	16.63969	0.399409	0.159528	0.569422	0.50	Non-signi

Table 13. The statistical parameters employed in the QSPR model of F.

PP								
	<i>N</i>	<i>A</i>	<i>b</i>	<i>r</i>	<i>r</i> ²	<i>F</i>	<i>p</i>	Indicator
BP	5	770.3666	−0.03048	0.111651	0.012466	0.03787	0.85	Non-signi
MW	5	345.0425	0.123694	0.447685	0.200422	0.751978	0.44	Non-signi
FP	5	419.2627	−0.01813	0.110239	0.012153	0.036906	0.85	Non-signi
MV	5	170.1161	0.154089	0.773686	0.598591	4.473665	0.12	Non-signi
C	5	799.7765	0.027259	0.46966	0.002206	0.006632	0.94	Non-signi

Table 14. Standard error of estimate.

TI	SE for BP	SE for MW	SE for FP	SE for MV	SE for C
ABC(G)	95.06779	71.46454	57.28143	26.34908	196.0973
RA(G)	93.91229	66.39218	56.58081	25.50293	191.58081
S(G)	93.77789	60.11675	56.49938	25.17546	191.1775
GA(G)	93.61597	65.74071	56.40126	24.97742	190.6539
M ₁ (G)	95.09238	71.85135	57.29593	27.5326	196.0216
M ₂ (G)	94.61798	69.88526	57.00796	26.97185	194.1227
HM(G)	95.36421	73.37161	57.4608	28.93528	197.0539
F(G)	95.60668	87.77978	57.62488	44.47212	204.3307
H(G)	92.73501	62.52702	55.86791	25.25247	187.5317

Table 15. Comparison of actual and computed values for Molar volume from regression models.

Name of drug	MV of drug	MV for ABC	MV for RA	MV for S	MV for GA	MV for M ₁	MV for M ₂	MV for F	MV for H	MV for HM
Glasdegib	281 cm ³	288.048	286.083	285.047	284.133	123.441	285.983	310.337	284.184	289.025
Palbociclib	340.7 cm ³	346.672	340.261	341.466	342.347	137.038	347.083	357.796	337.405	349.500
Daunorubicin	339.4 cm ³	363.980	369.134	368.851	368.860	141.017	367.257	368.582	372.219	365.782
Gilteritinib	444.9 cm ³	444.623	442.710	442.428	442.007	158.593	442.191	430.218	440.379	442.538
Ivosidenib	383.6 cm ³	346.980	351.805	368.851	352.250	136.374	347.083	322.664	355.412	342.755

Molecular weight = $98.01885 + 15.41259$ [H (G)]

Flash Point = $315.4363 + 3.156817$ [H (G)]

Molar Volume = $12.60989 + 13.33862$ [H (G)]

Complexity = $402.2318 + 16.63969$ [H (G)]

In the context of a quantitative structure analysis, a comparison between the correlation coefficients of the topological indices and physicochemical parameters is performed. Table 3 presents the physicochemical characteristics of drugs used in the treatment of blood cancer, while Table 2 shows the calculated values of the topological indices derived from their molecular structures. Furthermore, Table 4 provides the correlation coefficients between the topological indices and five different physicochemical characteristics. The relationship between the topological index and

Table 16. Comparison of actual and computed values for flash point from regression models.

Name of drug	FP of drug	FP for ABC	FP for RA	FP for S	FP for GA	FP for M ₁	FP for M ₂	FP for F	FP for H	FP for HM
Glasdegib	336.9 °C	387.535	383.261	382.689	382.053	387.537	385.434	402.764	379.709	388.833
Palbociclib	384.1 °C	395.610	393.743	393.892	393.971	395.791	395.393	397.180	392.304	396.142
Daunorubicin	4199.5 °C	397.994	399.329	399.330	399.399	398.206	398.681	395.911	400.544	398.110
Gilteritnib	375.3 °C	409.103	413.565	413.941	414.375	408.876	410.895	388.659	416.675	407.386
Ivosidenib	470 °C	395.555	395.900	395.945	395.999	395.388	395.393	401.314	396.566	395.327

Table 17. Comparison of actual and computed values for molecular weight from regression models.

Name of drug	Molar weight of drug	MW for ABC	MW for RA	MW for S	MW for GA	MW for M ₁	MW for M ₂	MW for F	MW for H	MW for HM
Glasdegib	374.4 (gmol ⁻¹)	424.742	4438.00	416.319	415.018	424.447	420.493	458.514	411.819	426.762
Palbociclib	447.5 (gmol ⁻¹)	485.384	5367.48	478.799	479.713	486.695	485.492	496.919	473.315	488.433
Daunorubicin	527.5 (g mol ⁻¹)	503.289	5862.82	509.178	509.178	504.914	506.953	505.648	513.542	505.037
Gilteritnib	552.7 (gmol ⁻¹)	586.710	7125.07	590.606	590.470	585.381	586.669	555.525	592.300	583.312
Ivosidenib	583.0 (g mol ⁻¹)	484.973	5558.73	490.248	490.719	483.659	485.492	468.489	494.122	481.554

Table 18. Comparison of actual and computed values for complexity from regression models.

Name of drug	Complexity of drug	C for ABC	C for RA	C for S	C for GA	C for M ₁	C for M ₂	C for F	C for H	C for HM
Glasdegib	595	770.477	754.259	752.074	749.623	769.559	761.405	824.582	741.102	773.865
Palbociclib	775	822.935	813.644	814.728	815.403	823.974	822.214	832.977	807.494	825.773
Daunorubicin	960	838.423	845.292	845.140	845.362	839.901	838.851	834.886	850.924	839.748
Gilteritnib	785	910.583	925.939	926.847	928.016	910.242	916.870	845.789	935.952	905.630
Ivosidenib	1050	822.579	825.863	826.593	826.593	821.320	822.214	826.762	829.958	819.983

Table 19. Comparison of actual and computed values for Boiling point from regression models.

Name of drug	BP of drug	BP for ABC	BP for RA	BP for S	BP for GA	BP for M ₁	BP for M ₂	BP for F	BP for H	BP for HM
Glasdegib	633.4	717.358	710.269	709.321	708.268	717.367	713.882	742.629	704.378	719.521
Palbociclib	711.5	730.666	727.578	727.824	727.953	730.964	730.306	733.241	725.195	731.546
Danorbicin	770	734.595	736.802	736.985	736.919	734.944	735.729	731.108	738.812	734.783
Gilteritnib	696.9	752.902	760.308	760.934	761.654	752.520	755.873	718.916	756.473	750.045
Ivosidenib	854.3	730.576	731.140	731.214	731.302	730.301	730.306	740.191	732.239	730.204

the physicochemical parameters of the drugs, along with their corresponding correlation coefficients, is illustrated in [Figure 2](#).

2.2. Computation of statistical parameters

We perform QSPR modeling to establish an association among the physicochemical characteristics of blood cancer drugs and their calculated degree-based topological indices (TIs). In the regression model, TIs represent the independent variable, *b* represents the model constant, *r* denotes the correlation coefficient, and *N* denotes the sample size. The highlighted theoretical and experimental calculations in the [Tables 5–13](#), indicate a high correlation coefficient. This type of test can aid in comparing different models and assessing their relative improvements. It should be highlighted that mostly the *p* amount is greater than 0.05 and the value of *r* is more than 0.6. Consequently, it determines that all attributes are important. [Tables 5–13](#) show the statistical

parameters. For the below tables following notations are important: BP will use for boiling points, MW for molecular weight, FP for flash point, MV for molecular volume, C for complexity and PP for physiochemical property.

2.3. Standard error of estimate (SE) and comparison

The SE, which is found in Table 14, is an indication of how much an observation differs from the estimated regression line. This table also provides an indication of the accuracy of the predictions made based on the regression line. To make further comparisons, Tables 15–19 contain both the experimentally and theoretically calculated values of the models in terms of their physicochemical properties.

3. Conclusions

The application of topological indices and statistical characteristics in linear QSPR models has demonstrated strong correlation values for various physicochemical properties of medications used to treat blood cancer. The results of this research offer valuable insights for the pharmaceutical sector in designing new drugs for the treatment of blood cancer. The significant impact of correlation coefficients among the variety of topological indices for these medications highlights the potential for estimating the physicochemical features of newly discovered anticancer drugs for the treatment of other specific cancer diseases. These findings are very useful for scientists studying pharmaceuticals, and provide a powerful tool for drug discovery and development. Here, we obtained that for molar volume, the ABC (G) index has a high correlation value of $r=0.92687$. The maximum complexity correlation offered by the HM index is $r=0.91108$. The GA index shows the highest correlation coefficient of $r=0.93454$ for the refractive index. The flash point's greatest correlation, $r=0.77368$, is provided by harmonic H (G).

For the medications used to treat blood cancer, we generated topological indices and compared them to a linear QSPR model in this work. The pharmaceutical sector will be able to design new treatments using the outcomes of this research to discover safety precautions for the mentioned disease. When it comes to the variety of topological indices for these medications, the correlation coefficient is a significant factor. The facts are eye-opener for scientists studying drugs in the pharmaceutical sector, and offer a method for estimating the physicochemical features of newly discovered anticancer drugs to treat other particular cancer diseases.

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Disclosure statement

The authors declare no any conflict of interest/competing interests.

Author contributions

All the authors have equally contributed to this manuscript in all stages, from conceptualization to the write-up of final draft.

Data availability statement

All data generated or analyzed during this study are included in this article.

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